Introducción al aprendizaje de máquina

Aprendizaje de Máquina Aplicado

2026

Yomin Jaramillo Múnera
yejaramilm@eafit.edu.co

Vigilada Mineducación

UNIVERSIDAD EAFIT

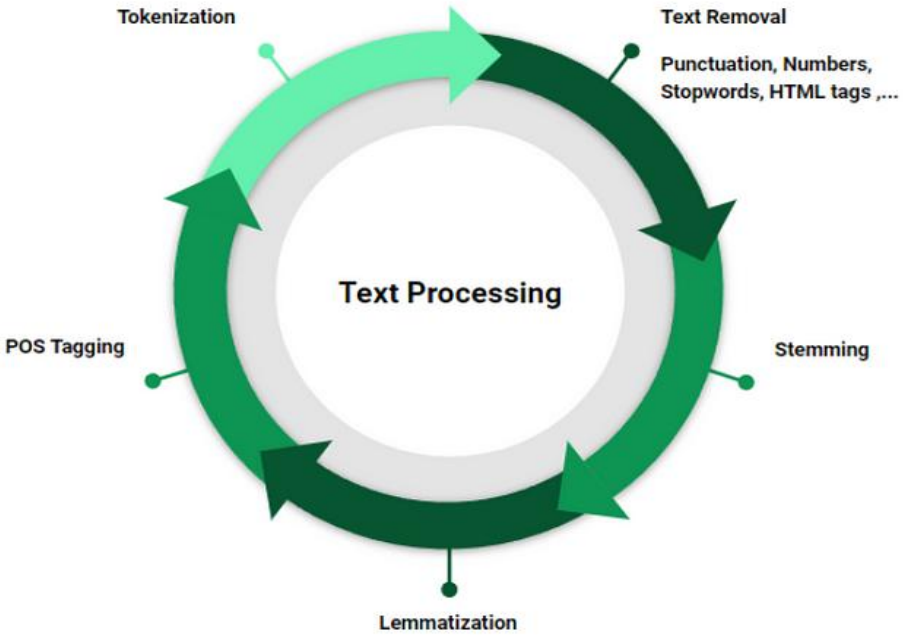
Agenda

- Introducción a NLP
- Tokenizacion
- Embebbings
- Autoregresive models
- Self-attention
- Mul-tihead Attention
- Transformers
- Slef-supervised learning


ChatGPT 4
How can I help you today?
Message ChatGPT...
ChatGPT can make mistakes. Consider checking important information.

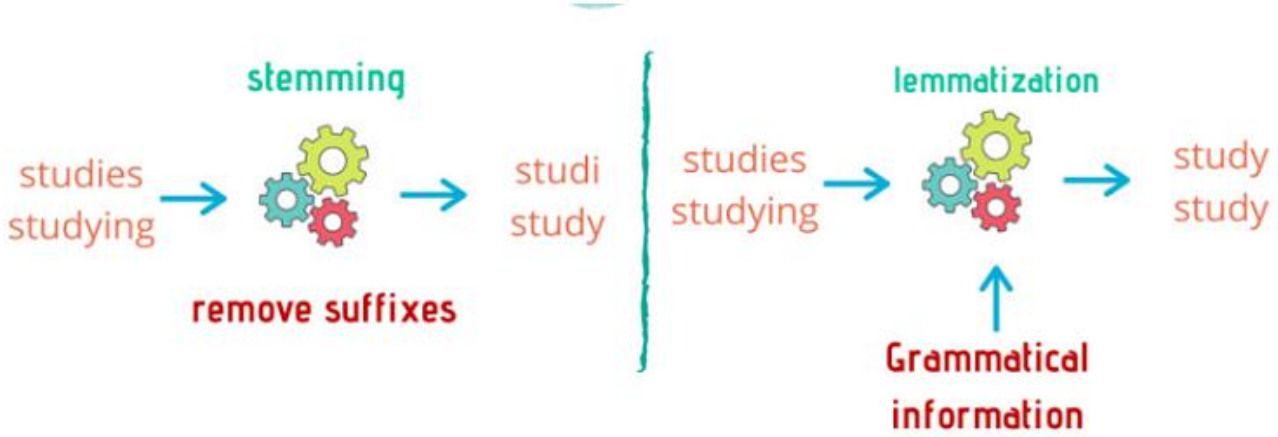

Untitled
What can I help you with today?
Message Claude...
Claude 3 Opus

Introducción a NLP

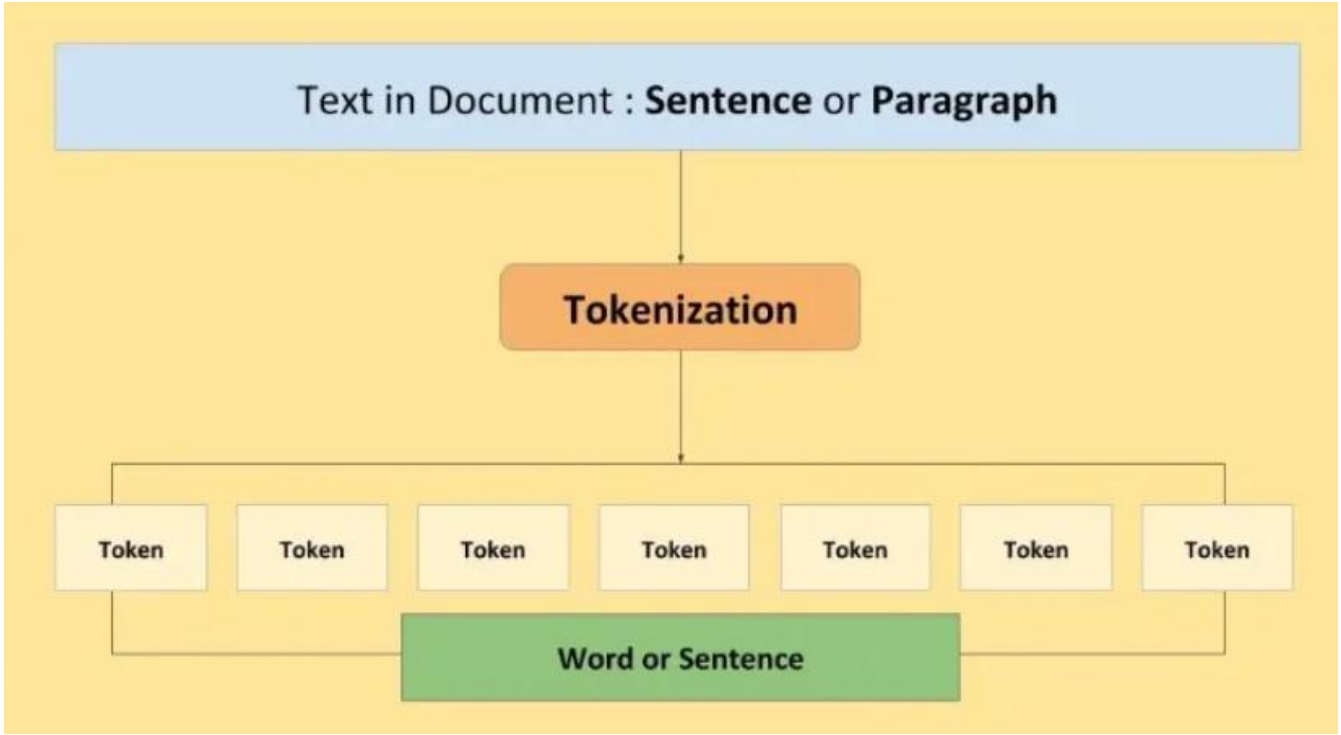


El NLP (**Natural Language Processing o Procesamiento del Lenguaje Natura**l) es una rama de la inteligencia artificial que permite a las máquinas comprender, interpretar y generar lenguaje humano.

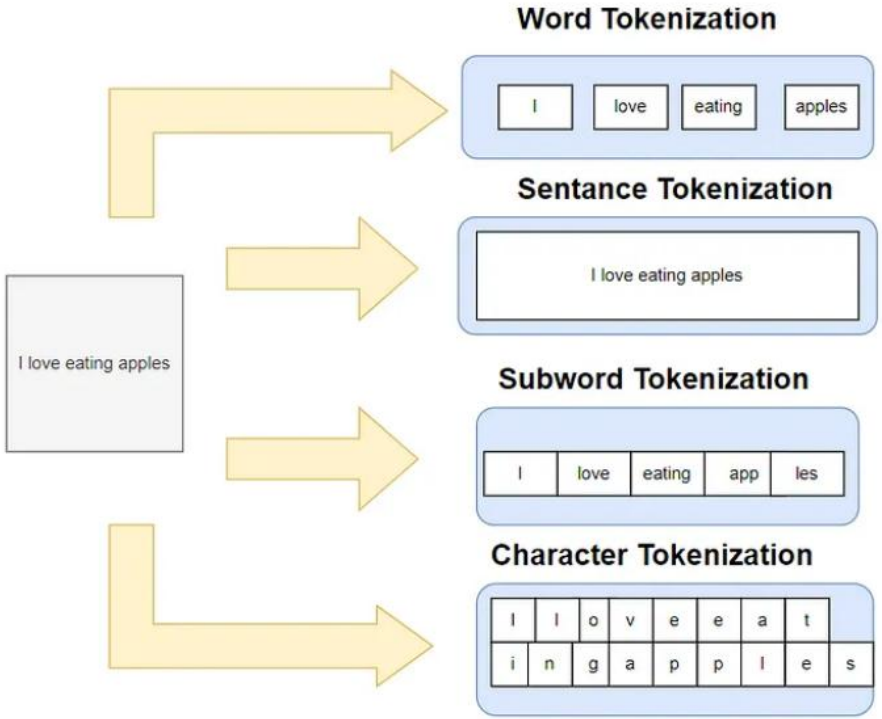
Stemming and Lemmatization



Tokenization

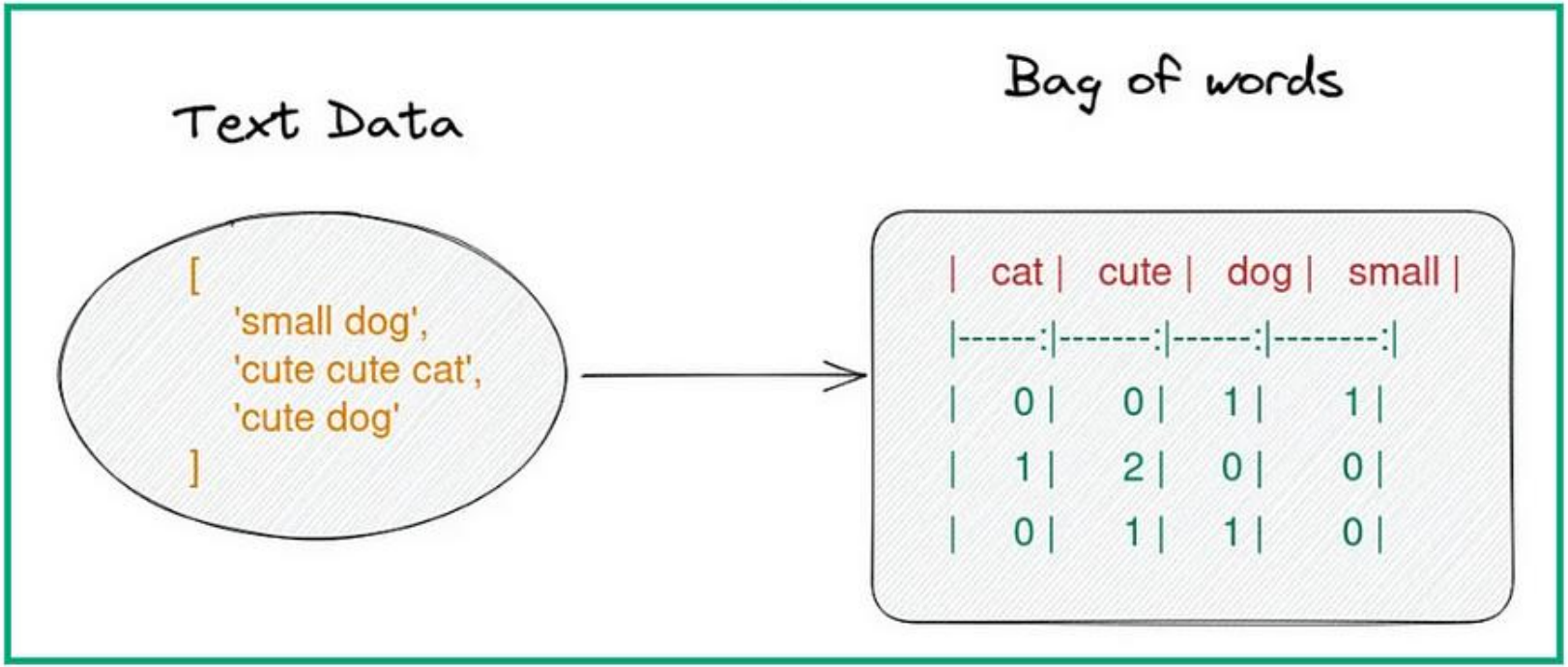


Tokenization



Tokenization techniques

Bag of words


Text Data
Bag of words
[
'small dog',
'cute cute cat',
'cute dog'
]
cat	cute	dog	small
0	0	1	1
1	2	0	0
0	1	1	0

Hello Transformers

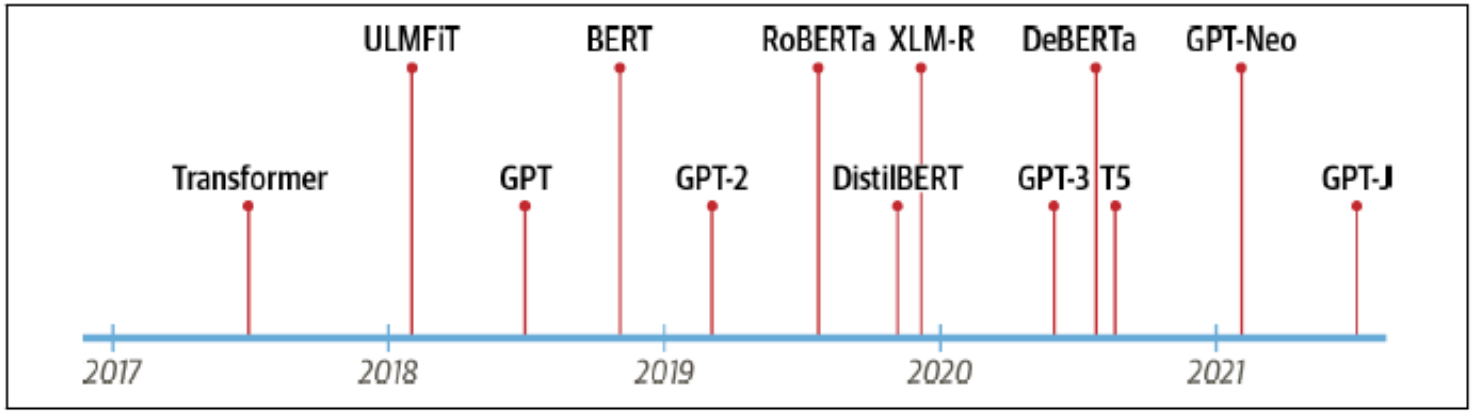
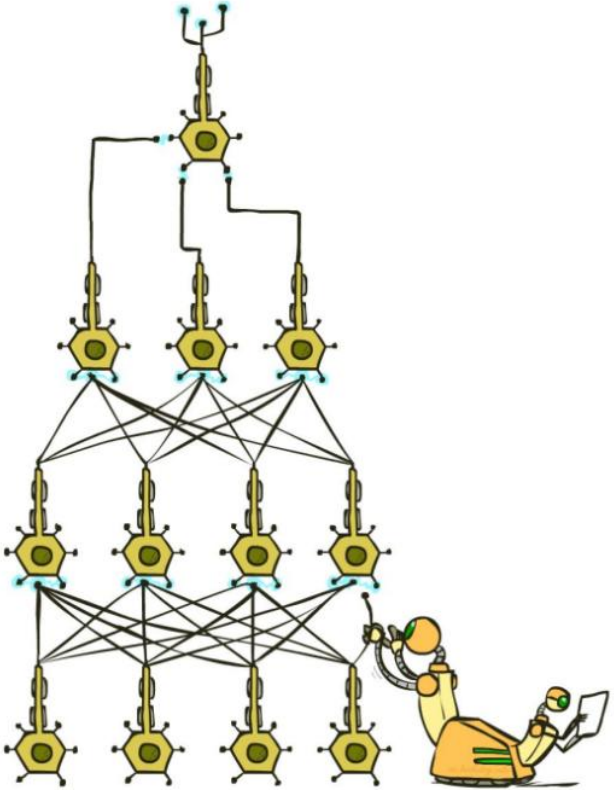


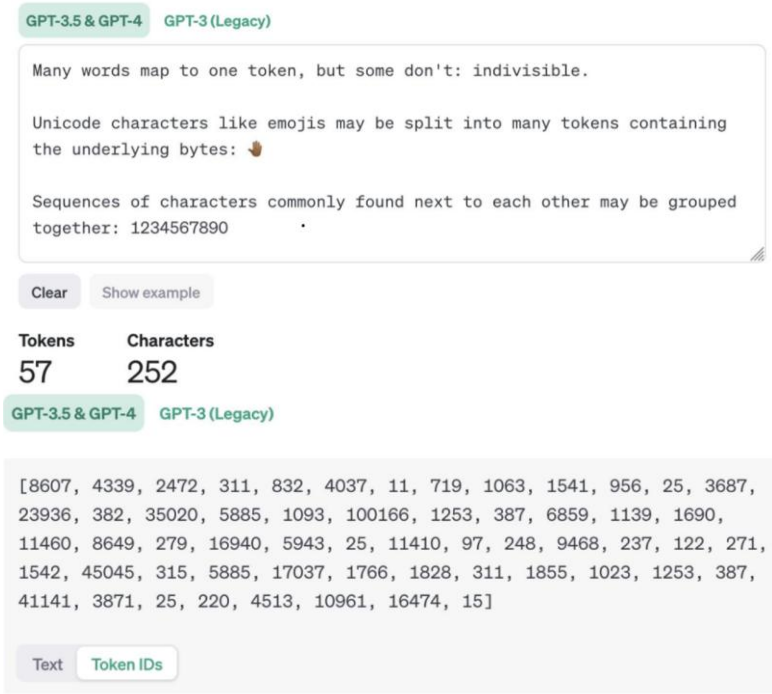
Figure 1-1. The transformers timeline

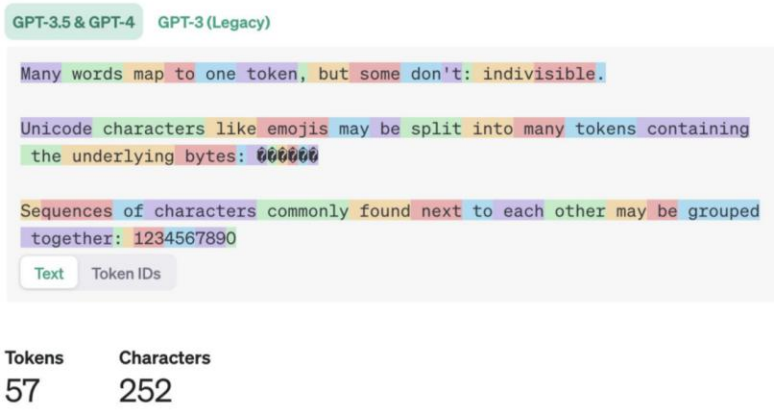
LLMs



- Input: some text
  - “The dog chased the”
- Output: more text
  - … “ ball”
- Implementation:
  - Linear algebra
  - How??

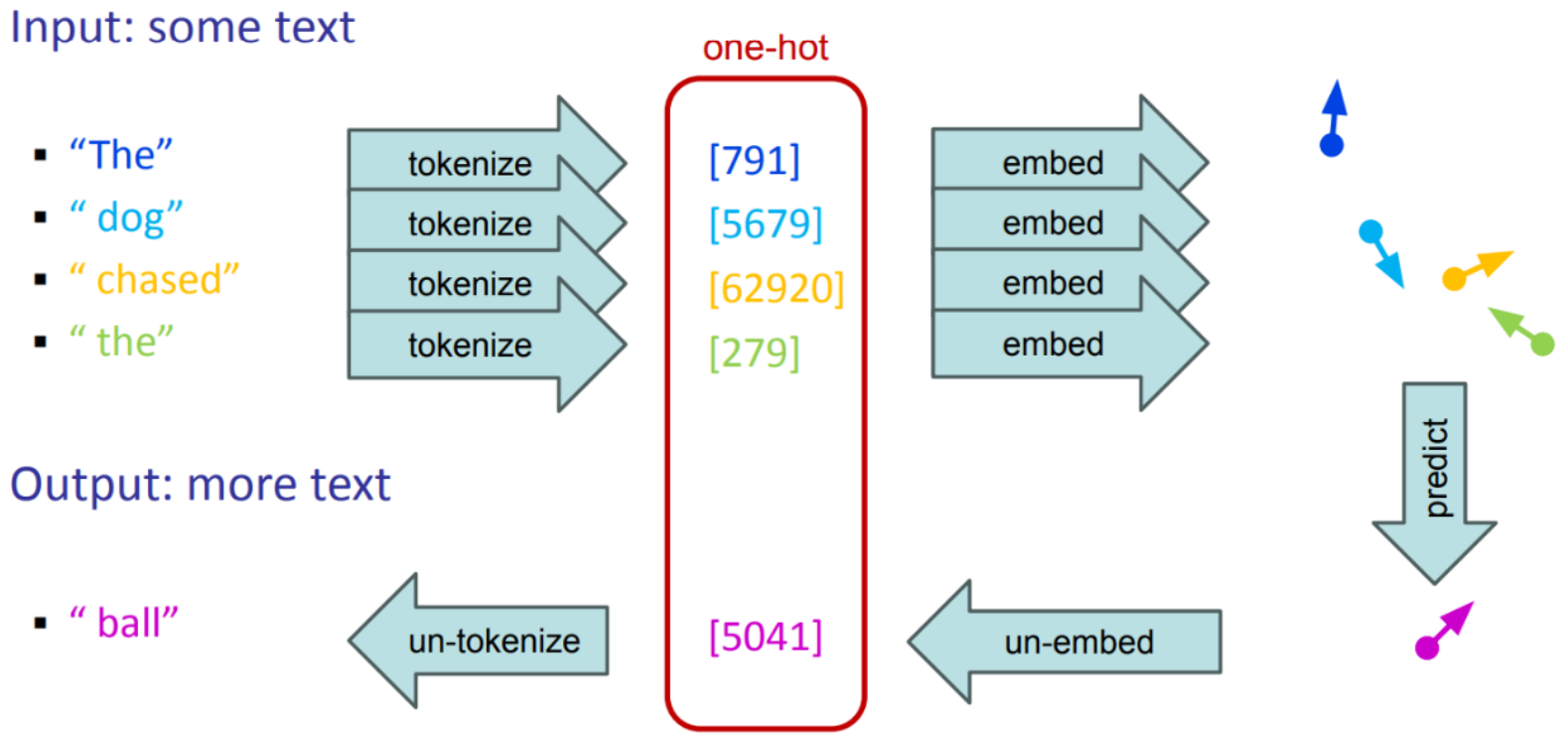
Tokenization



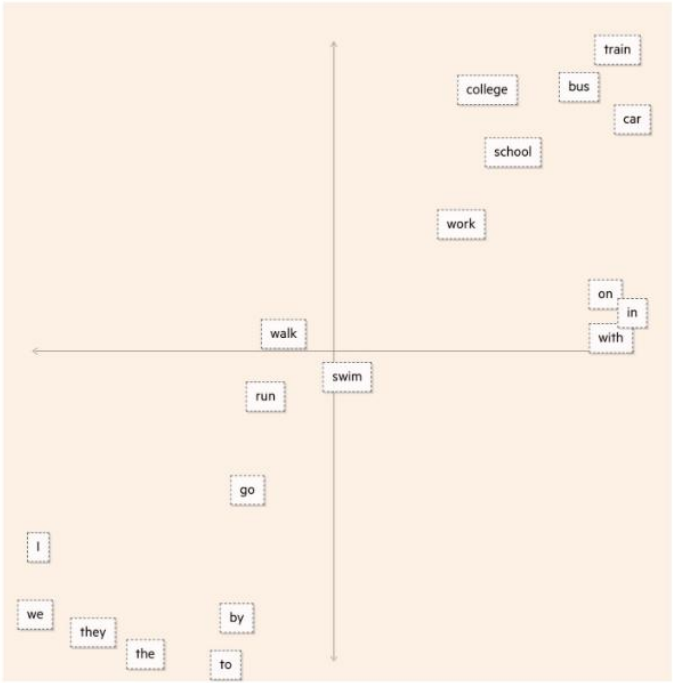


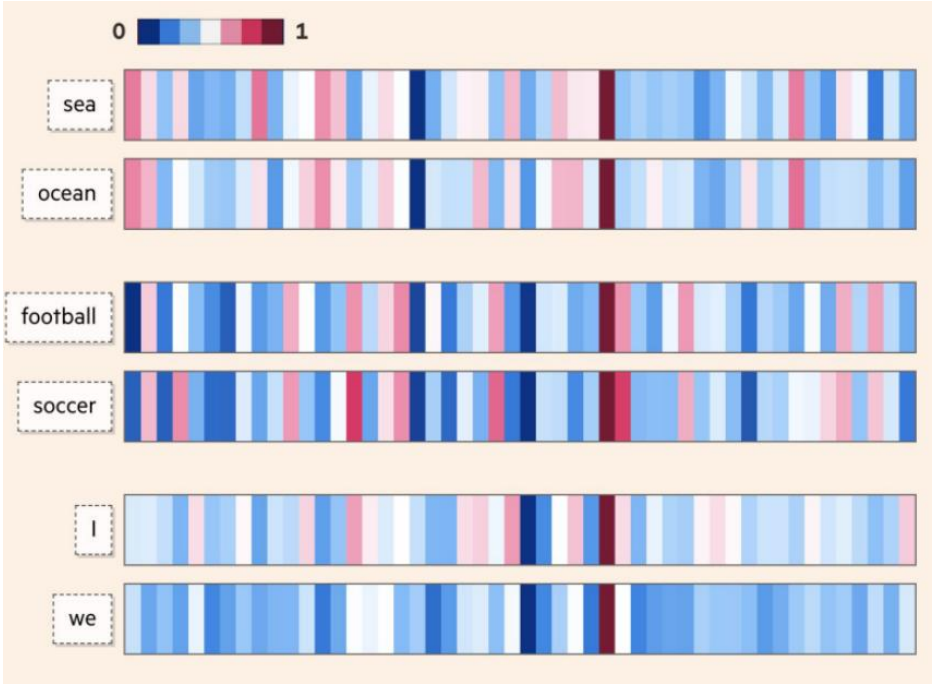
https://gptforwork.com/tools/tokenizer

Embebbing


Input: some text
“The”
“ dog”
“ chased”
“ the”
tokenize
tokenize
tokenize
tokenize
one-hot
[791]
[5679]
[62920]
[279]
embed
embed
embed
embed
predict
Output: more text
“ ball”
un-tokenize
[5041]
un-embed

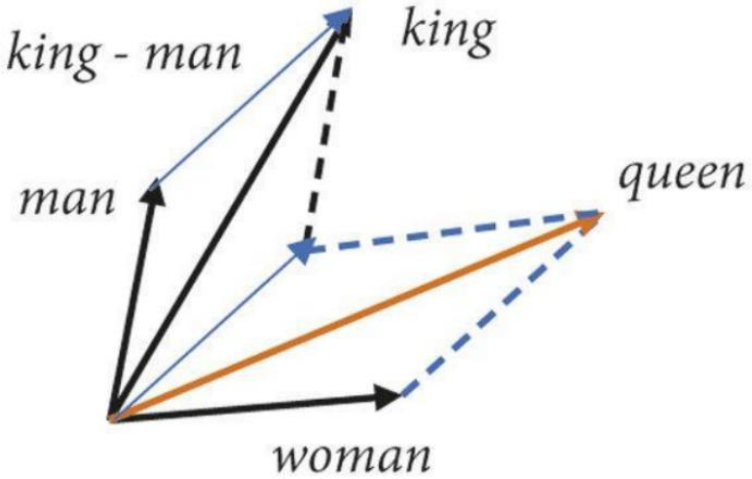
Embebbing





https://projector.tensorflow.org/

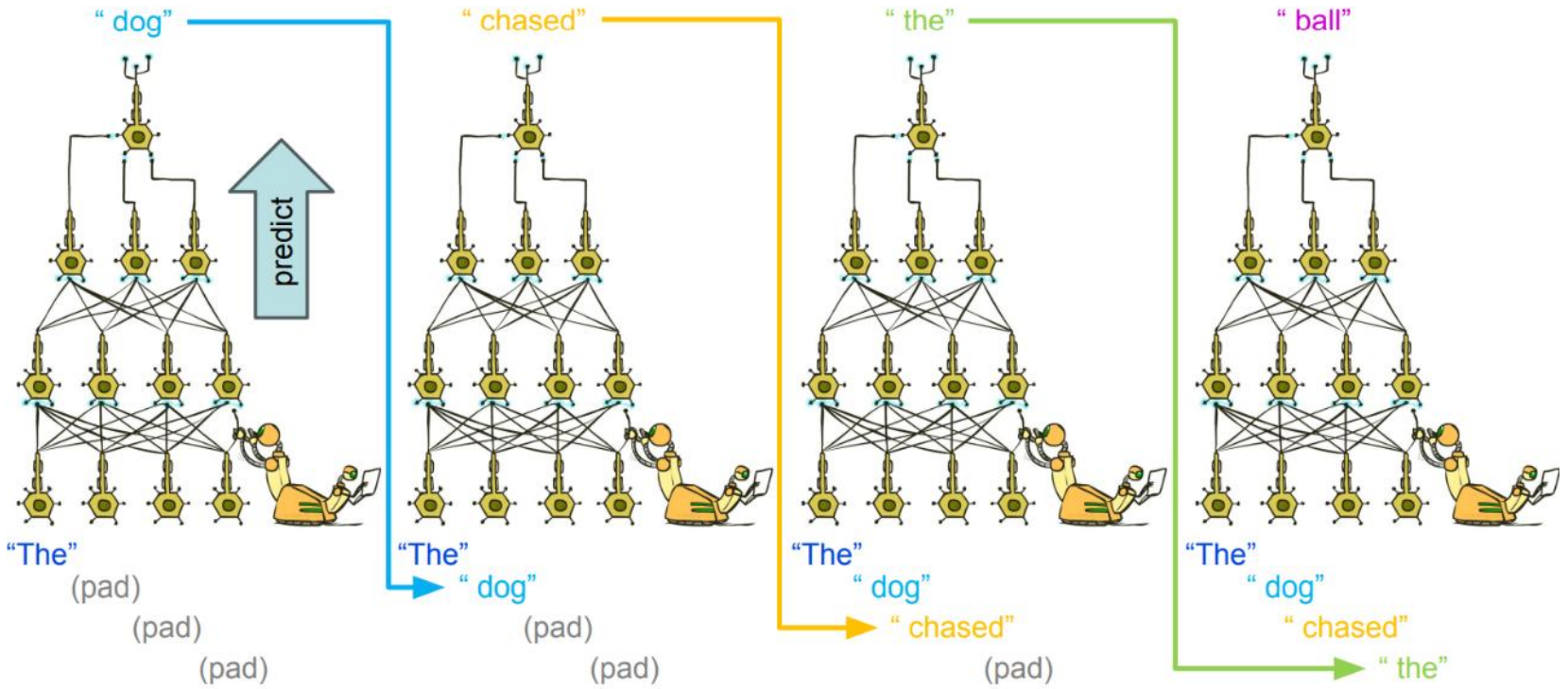
Second step: Embebbing



$$king - man + woman \approx queen$$

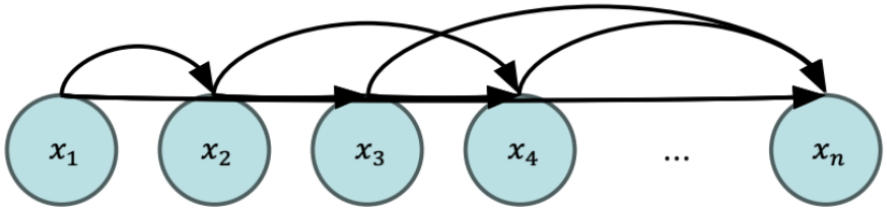
[Efficient estimation of word representations in vector space, Mikolov et al, 2013]

Autoregressive models

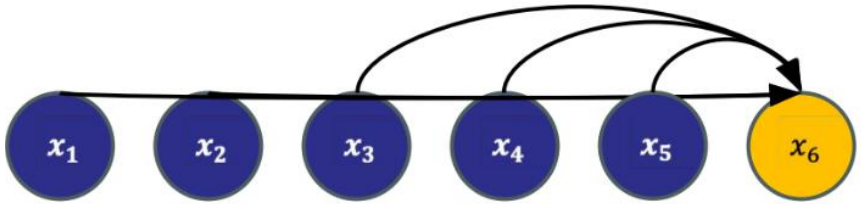


Autoregressive models

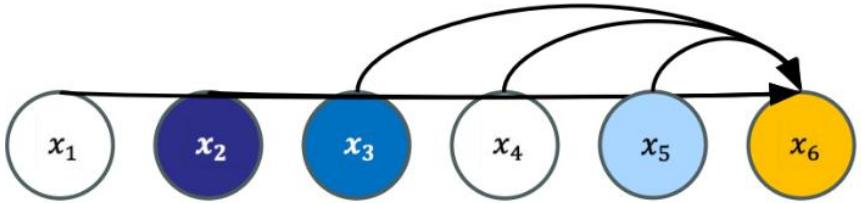
- Predict output one piece at a time (e.g. word, token, pixel, etc.)
- Concatenate: input + output
- Feed result back in as new input
- Repeat



Self –attention mechanism

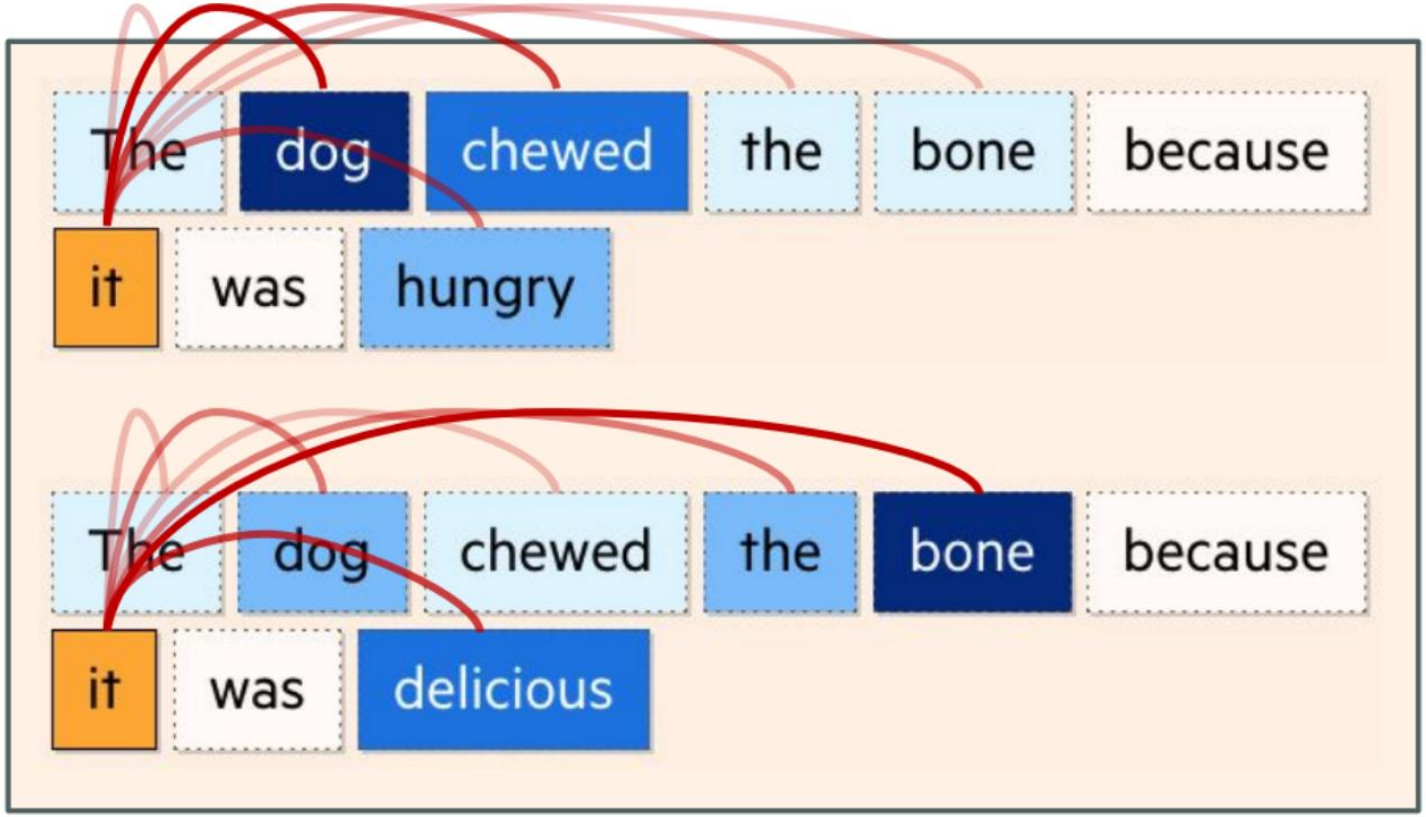


- Instead of conditioning on *all* input tokens equally…

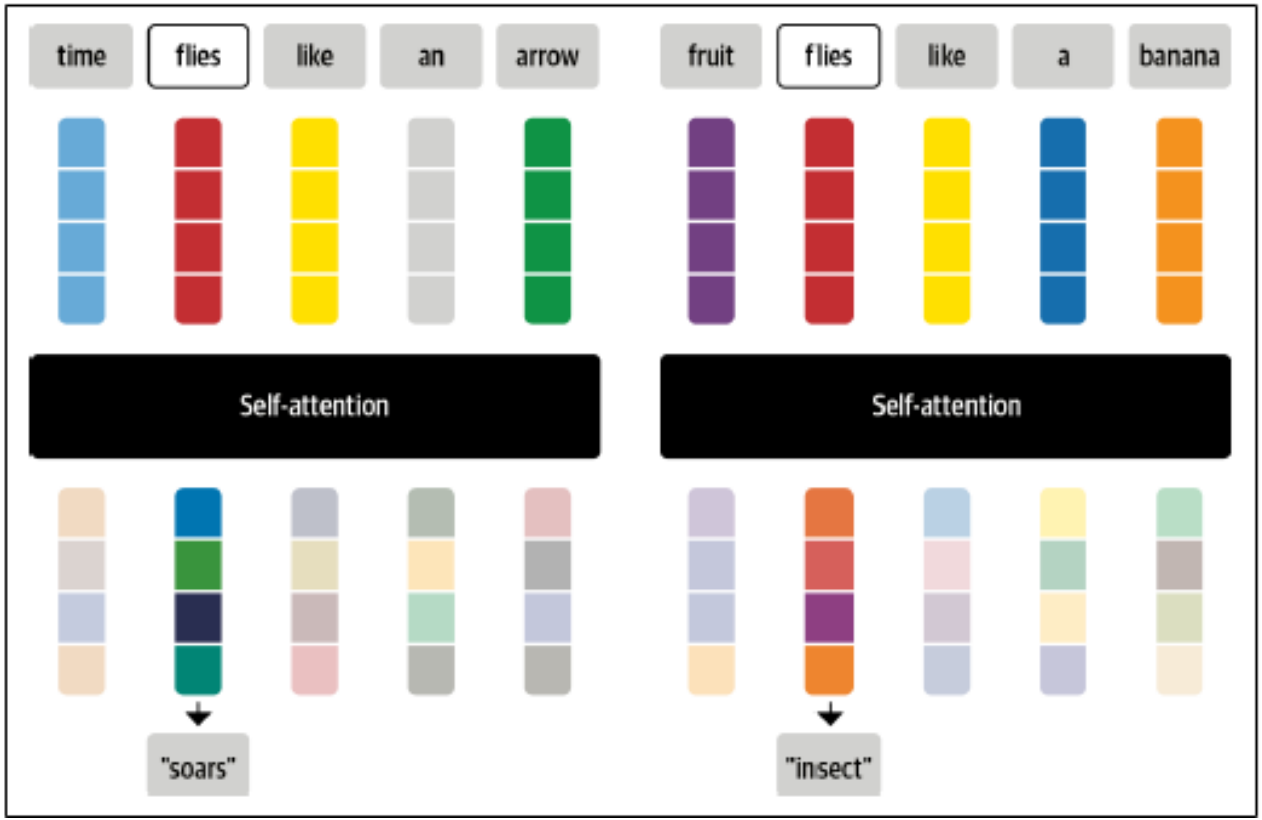


- Pay more attention to relevant tokens!

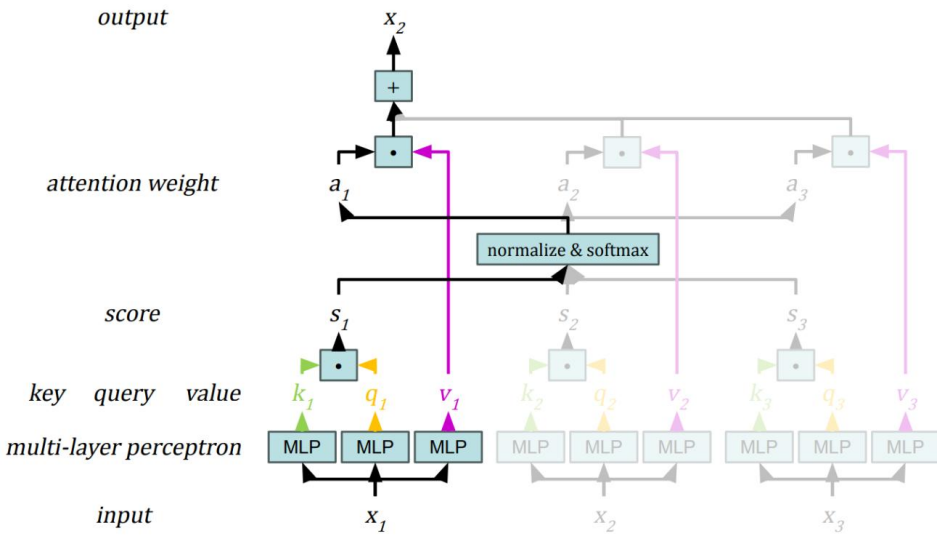
Self –attention mechanism

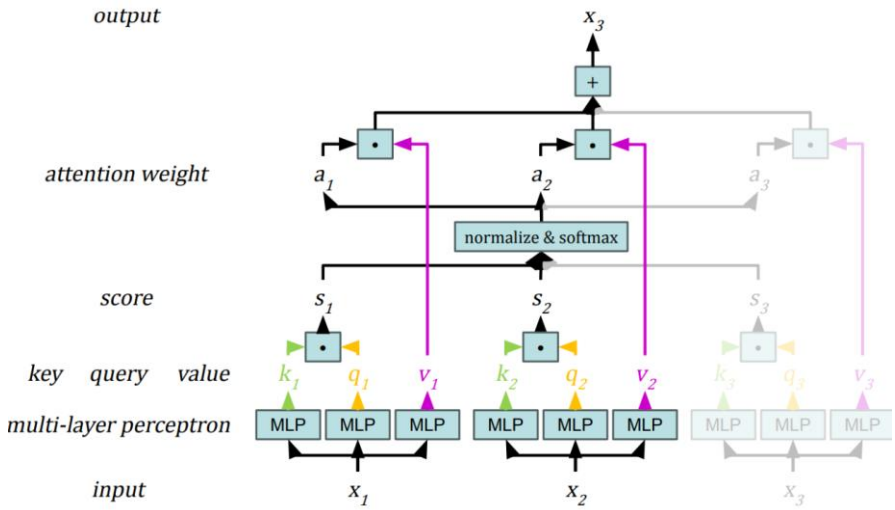


Self-Attention

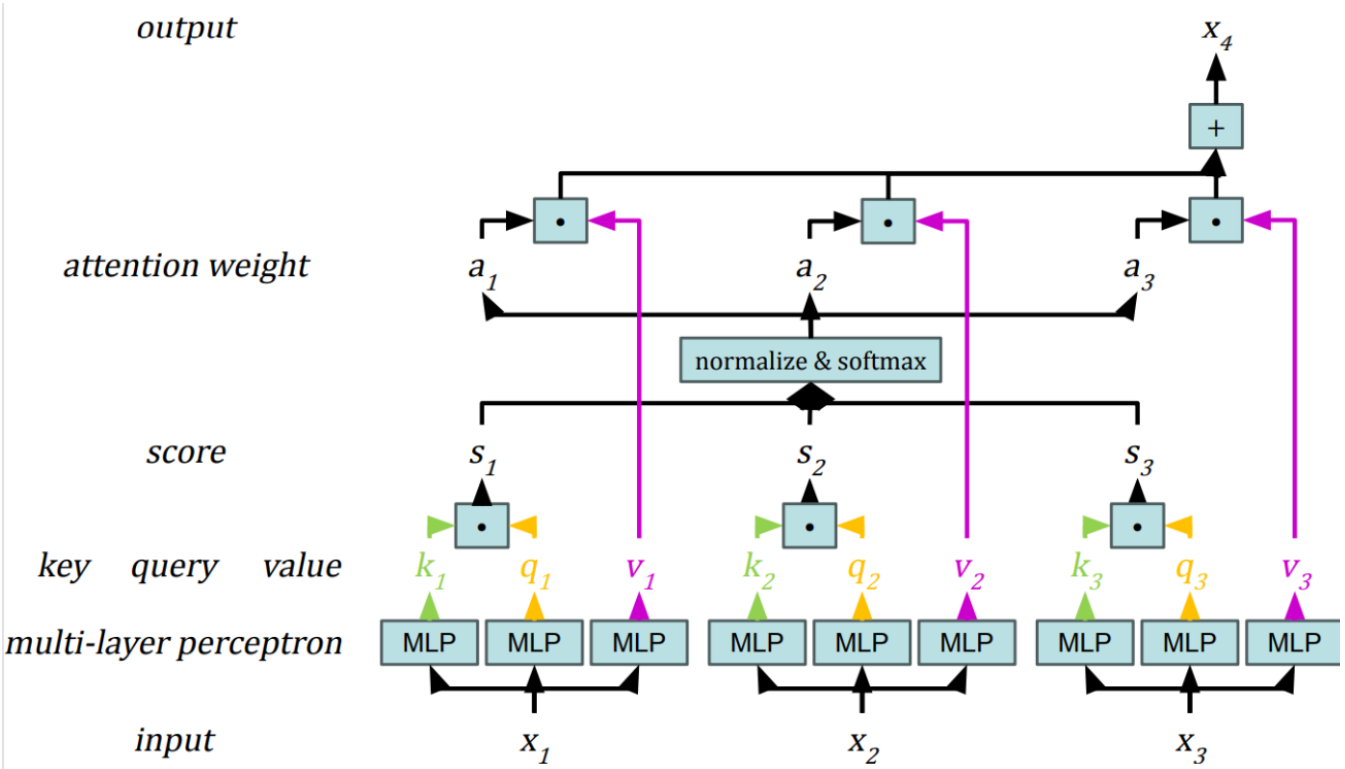


Self –attention mechanism



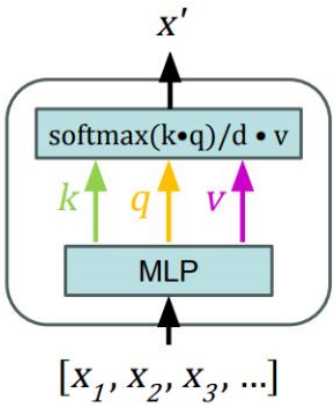


Self –attention mechanism

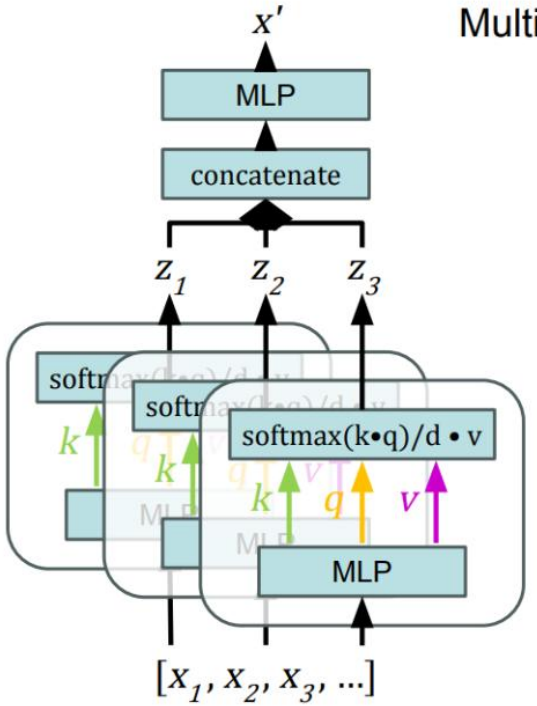


Multihead attention mechanism

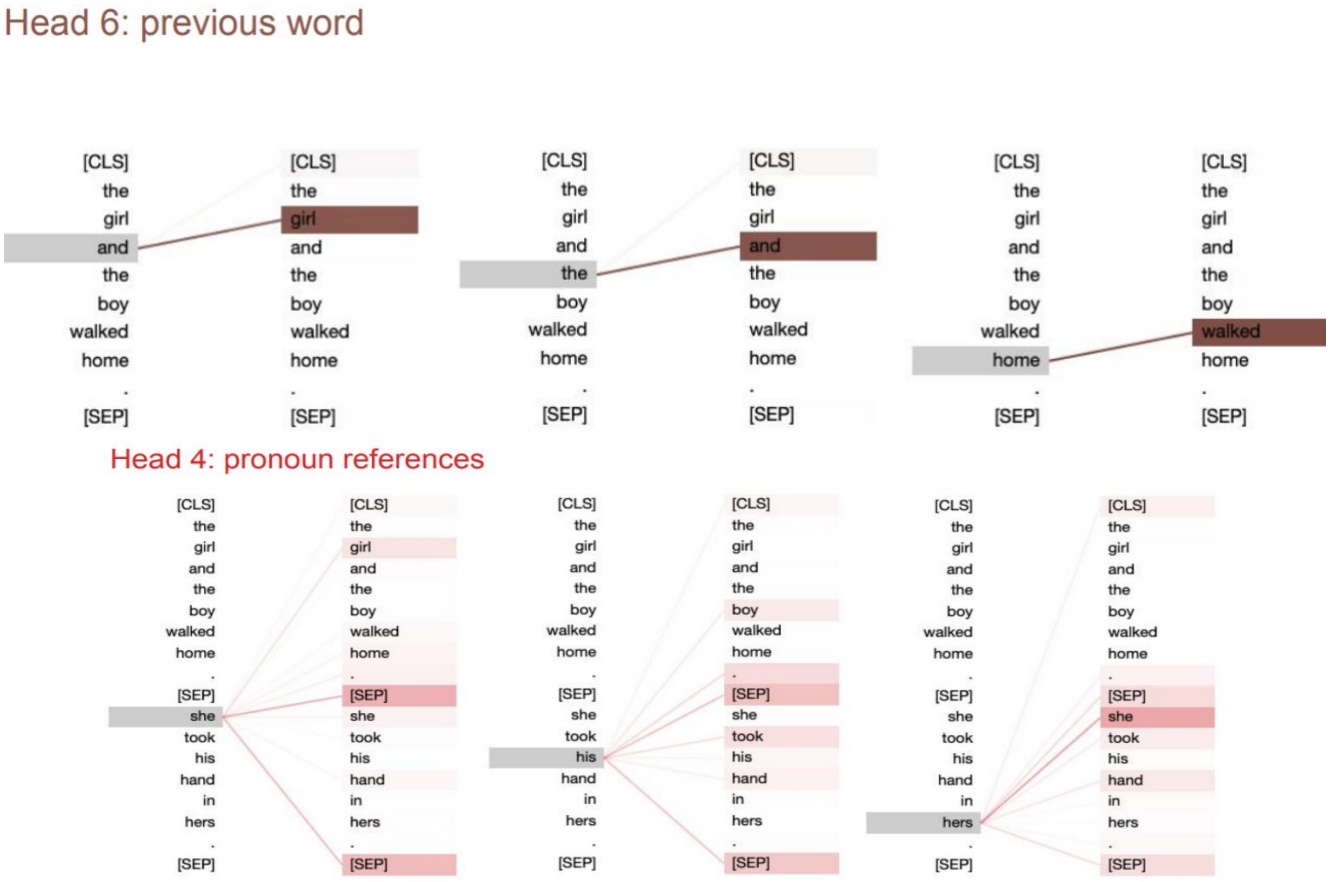
Single-headed



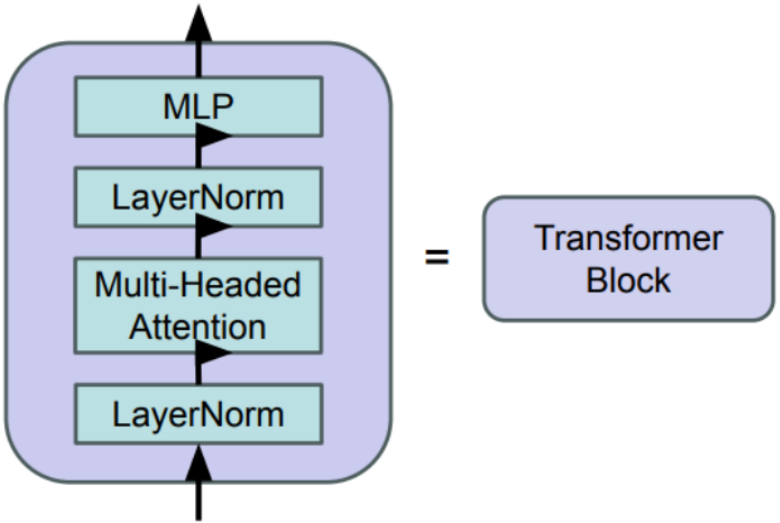
Multi-headed

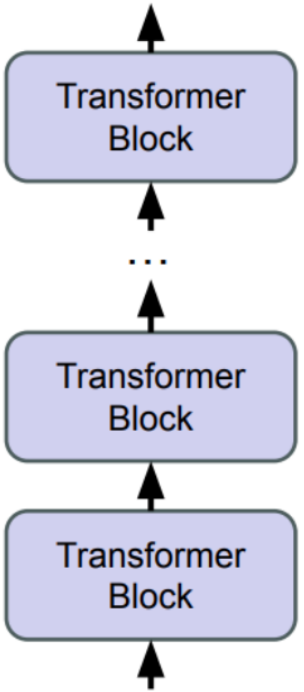


Multihead attention mechanism

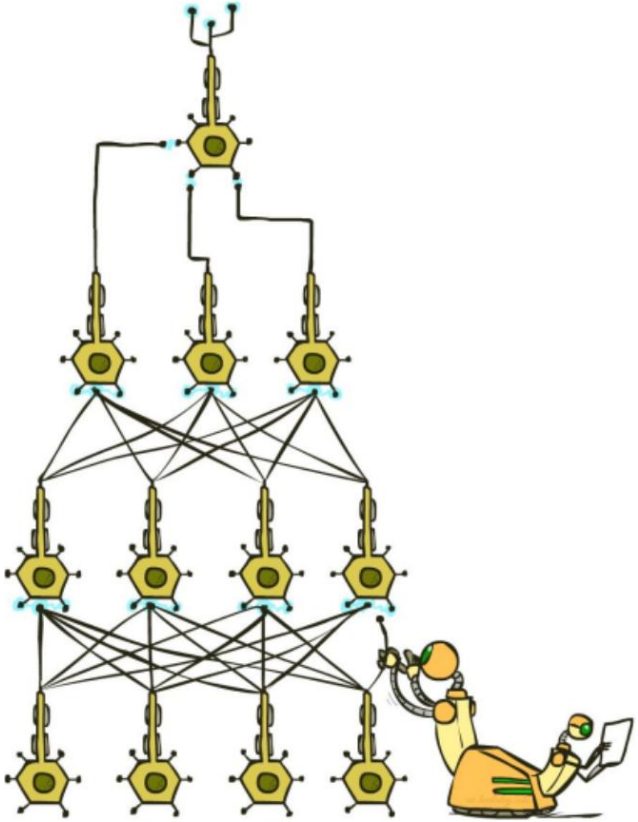


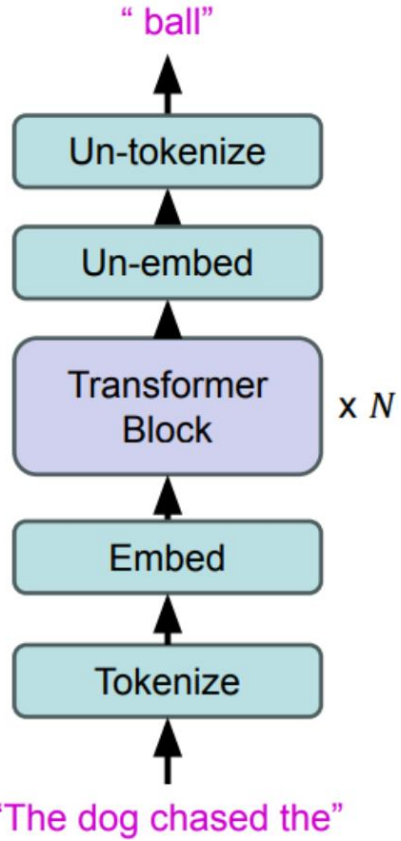
Transformer block





Transformer block





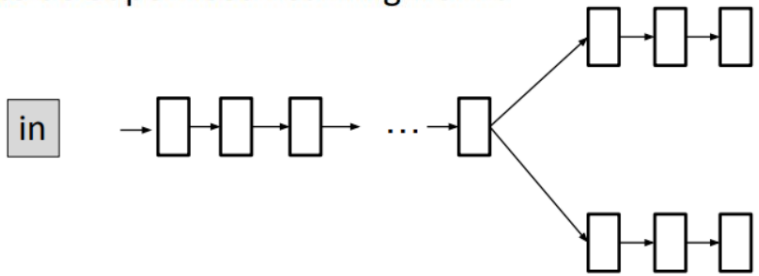
Self-Supervised Learning

- Do we always need human supervision to learn features?
- Can't we learn general-purpose features?
- Key hypothesis:

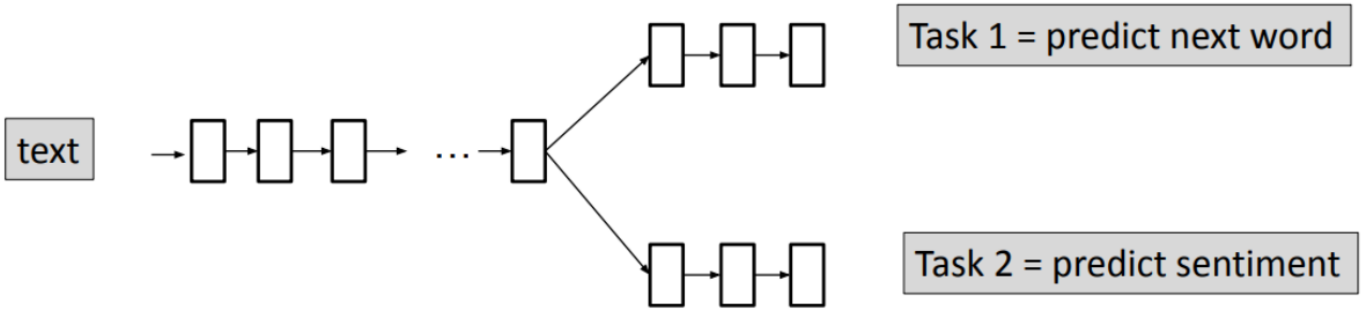
Task 1 IF neural network smart enough to predict:

- Next frame in video
- Next word in sentence
- Generate realistic images
- ``Translate'' images
- …

Task 2 THEN same neural network is ready to do Supervised Learning from a very small data-set



Self-Supervised Learning







Fine-tuning

1. **Pre-Train:** train a large model with a lot of data on a self-supervised task
   - Predict next word / patch of image
   - Predict missing word / patch of image
   - Predict if two images are related (contrastive learning)
2. **Fine-Tune:** continue training the same model on task you care about

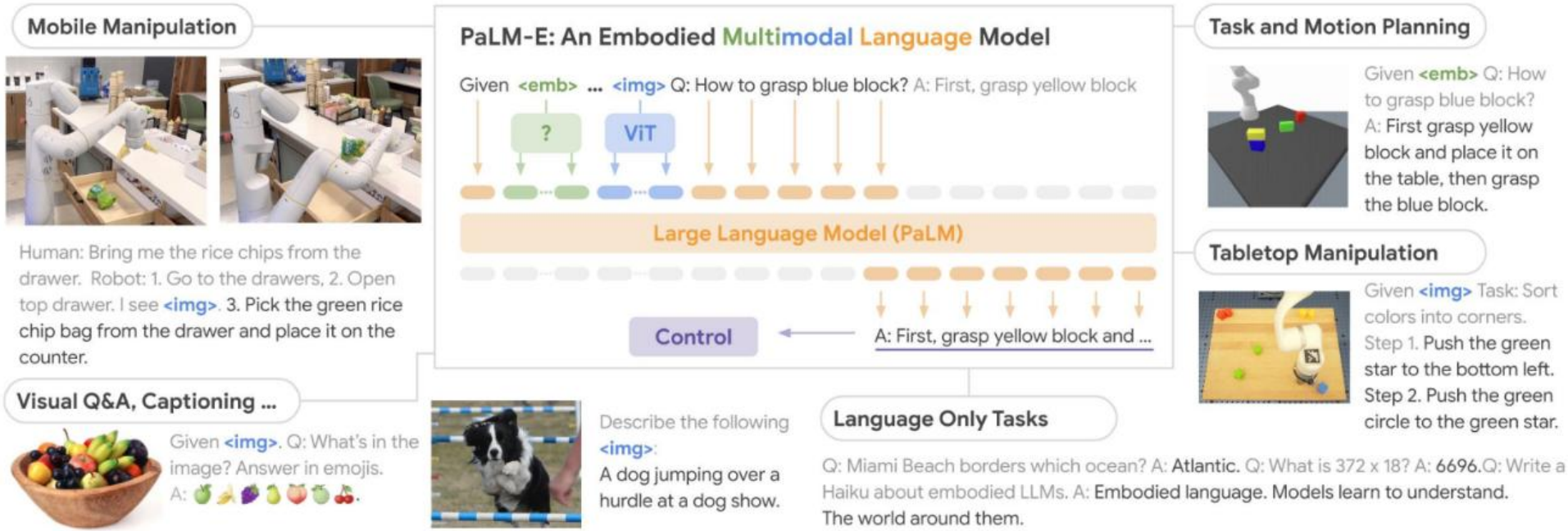
Fine-tuning

- Task 1 = predict next word (learns to mimic human-written text)
  - Query: "What is population of Berkeley?"
  - Human-like completion: "This question always fascinated me!"

- Task 2 = generate **helpful** text
  - Query: "What is population of Berkeley?"
  - Helpful completion: "It is 117,145 as of 2021 census."

- Fine-tune on collected examples of helpful human conversations
- Also can use Reinforcement Learning

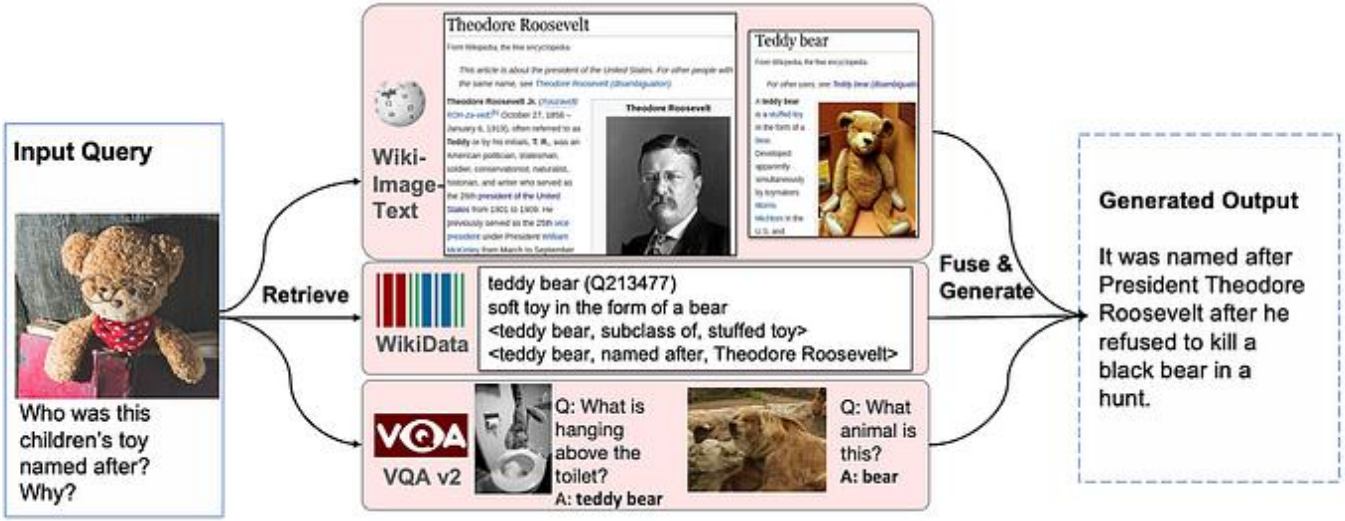
Currently


Mobile Manipulation
Human: Bring me the rice chips from the drawer. Robot: 1. Go to the drawers, 2. Open top drawer. I see <img>. 3. Pick the green rice chip bag from the drawer and place it on the counter.
Visual Q&A, Captioning ...
Given <img>. Q: What's in the image? Answer in emojis.
A:
PaLM-E: An Embodied Multimodal Language Model
Given <emb> ... <img> Q: How to grasp blue block? A: First, grasp yellow block
?
ViT
Large Language Model (PaLM)
Control
A: First, grasp yellow block and ...
Describe the following <img>:
A dog jumping over a hurdle at a dog show.
Task and Motion Planning
Given <emb> Q: How to grasp blue block? A: First grasp yellow block and place it on the table, then grasp the blue block.
Tabletop Manipulation
Given <img> Task: Sort colors into corners. Step 1. Push the green star to the bottom left. Step 2. Push the green circle to the green star.
Language Only Tasks
Q: Miami Beach borders which ocean? A: Atlantic. Q: What is 372 x 18? A: 6696.Q: Write a Haiku about embodied LLMs. A: Embodied language. Models learn to understand. The world around them.

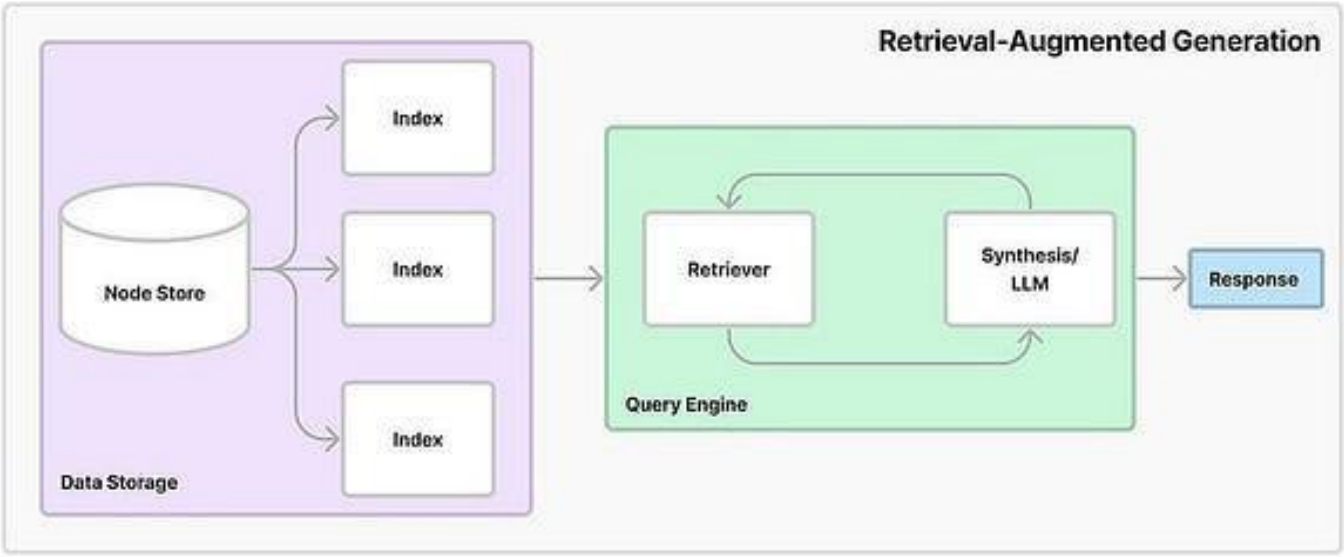
RAG

Retrieval-Augmented Generation (RAG)

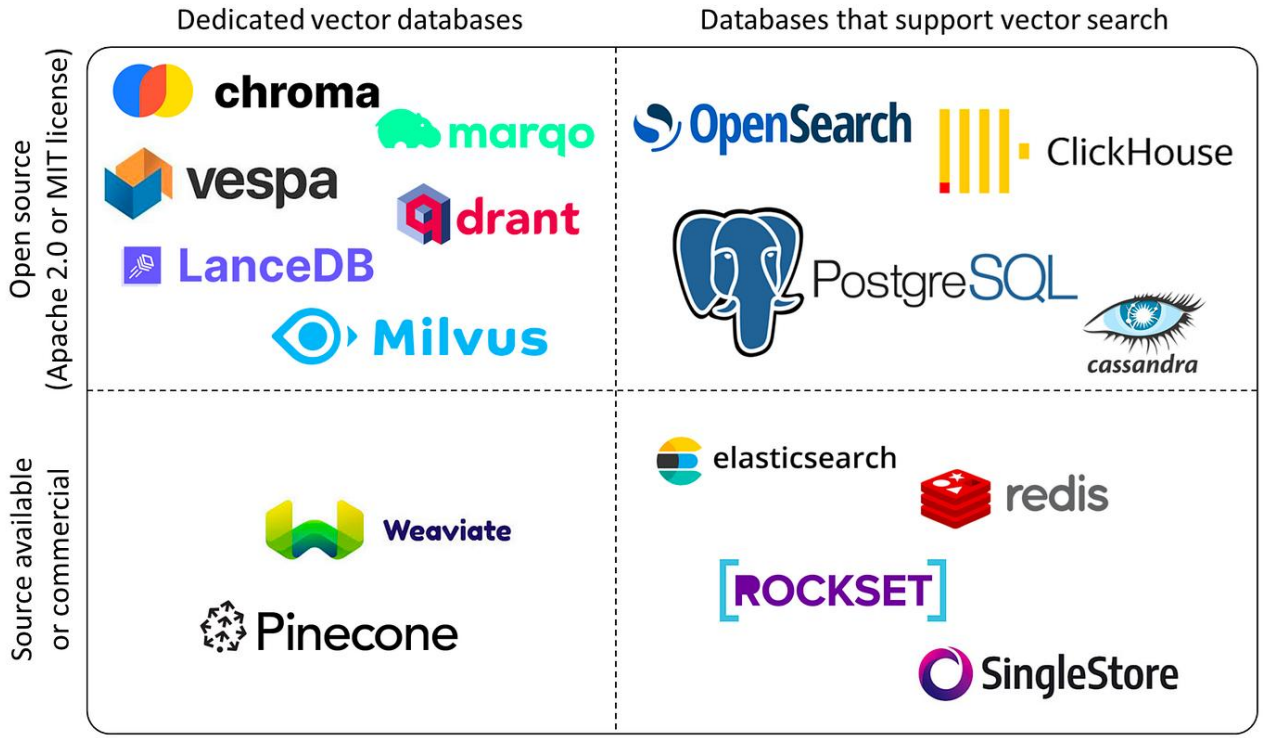
Retrieval-augmented generation (RAG) is a natural language processing (NLP) approach that combines the benefits of both retrieval-based and generation-based methods for content generation tasks. It aims to improve the quality and controllability of the generation tasks by leveraging a pre-trained language model in conjunction with a retrieval mechanism.



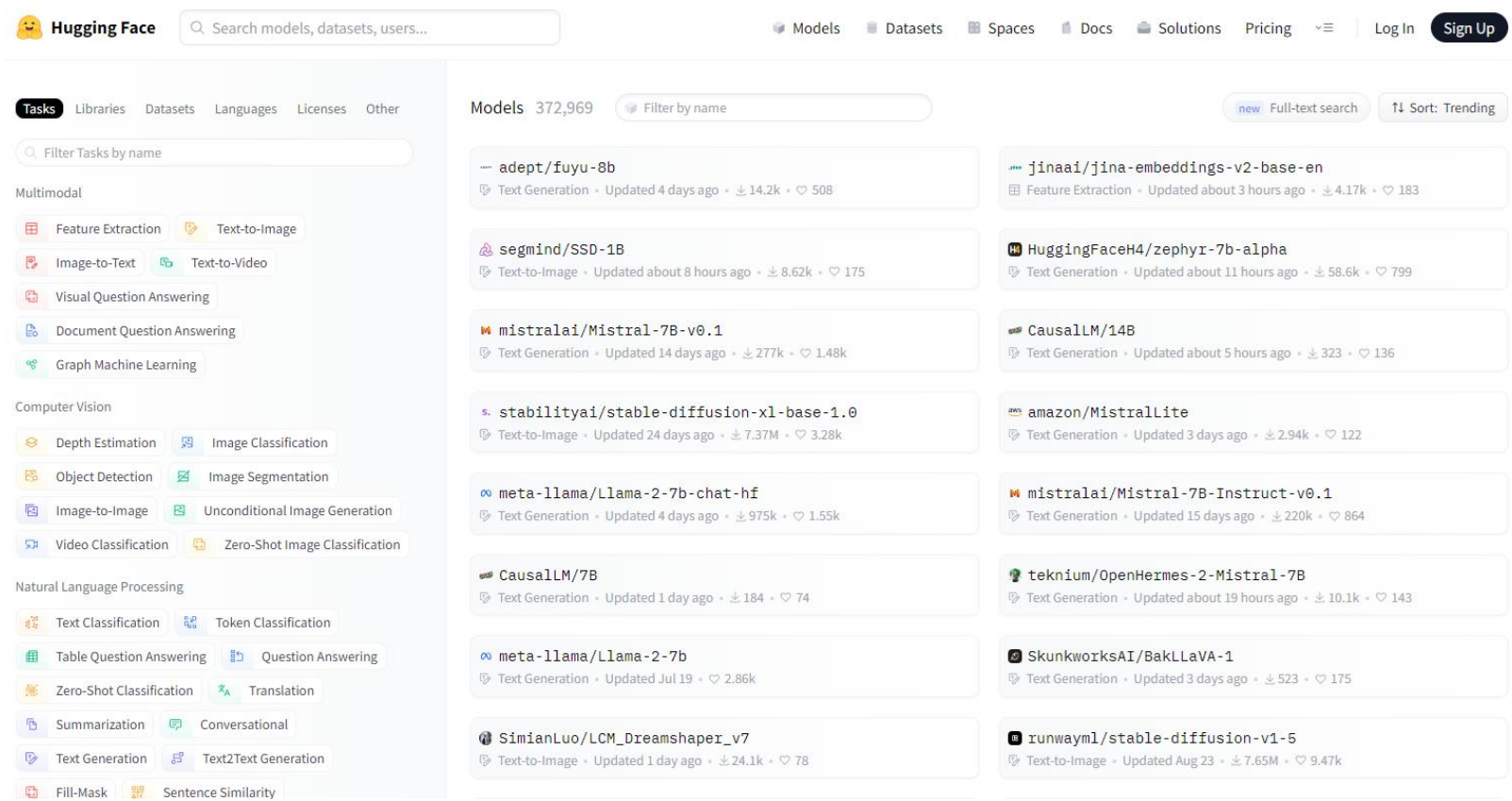
RAG: Componentes



RAG: Database



Hugging Face Hub



RAG: LLM



Hugging Face

The task illustrated in this tutorial is supported by the following model architectures:

ALBERT, BART, BERT, BigBird, BigBird-Pegasus, BLOOM, CamemBERT, CANINE, ConvBERT, Data2VecText, DeBERTa, DeBERTa-v2, DistilBERT, ELECTRA, ERNIE, ErnieM, Falcon, FlauBERT, FNet, Funnel Transformer, OpenAI GPT-2, GPT Neo, GPT NeoX, GPT-J, I-BERT, LayoutLMv2, LayoutLMv3, LED, LiLT, Longformer, LUKE, LXMERT, MarkupLM, mBART, MEGA, Megatron-BERT, MobileBERT, MPNet, MPT, MRA, MT5, MVP, Nezha, Nyströmformer, OPT, QDQBert, Reformer, RemBERT, RoBERTa, RoBERTa-PreLayerNorm, RoCBert, RoFormer, Splinter, SqueezeBERT, T5, UMT5, XLM, XLM-RoBERTa, XLM-RoBERTa-XL, XLNet, X-MOD, YOSO